🤖 AI (Artificial Intelligence) Roadmap (Week-wise Structured Plan)

🧠 PHASE 1: Foundations (Python + Math Basics)

📅 Weeks 1–6

📘 Week 1: Introduction to AI

- What is AI?
- Difference:
  - AI vs ML vs Deep Learning

👉 Setup:

- Install Python
- Use Jupyter Notebook / VS Code

📘 Week 2: Python Basics

- Variables
- Data types
- Lists, dictionaries

👉 Practice:

- Basic coding problems

📘 Week 3: Control Flow + Functions

- if/else
- loops
- functions

📓 Week 4: Math Basics (VERY IMPORTANT)

- Linear Algebra:
  - Vectors
  - Matrices

📓 Week 5: Statistics Basics

- Mean, median
- Probability basics

📓 Week 6: NumPy & Pandas

- Arrays
- Data handling

👉 Mini Task:

- Analyze a small dataset

🧠 PHASE 2: Machine Learning Basics

📅 Weeks 7–16

📓 Week 7: Intro to Machine Learning

- Types:
  - Supervised
  - Unsupervised

📓 Week 8: Linear Regression

- Concept + implementation

Week 9: Logistic Regression

- Classification

Week 10: Model Evaluation

- Accuracy
- Precision / Recall

Week 11: Decision Trees

- Tree-based models

Week 12: Random Forest

- Ensemble methods

Week 13: K-Nearest Neighbors

- Distance-based learning

Week 14: Clustering

- K-Means

Week 15: Feature Engineering

- Data preprocessing

Week 16: ML Project

Build:

- Prediction model (real dataset)

PHASE 3: Deep Learning

Weeks 17–28

Week 17: Neural Networks Basics

- Perceptron
- Layers

Week 18: Forward & Backpropagation

- How networks learn

Week 19: TensorFlow / PyTorch Setup

- Choose one framework

Week 20: Build First Neural Network

- Simple model

Week 21: CNN (Computer Vision)

- Image classification

📓 Week 22: CNN Project

👉 Build:

- Image classifier

📓 Week 23: RNN Basics

- Sequential data

📓 Week 24: LSTM

- Improved RNN

📓 Week 25: NLP Basics

- Text processing

📓 Week 26: NLP Project

👉 Build:

- Sentiment analysis

📓 Week 27: Model Tuning

- Hyperparameters

📓 Week 28: Deep Learning Project

👉 Build:

- End-to-end DL model

⍰ PHASE 4: Advanced AI (LLMs + Specialization)

📅 Weeks 29–36

📘 Week 29: Transformers Basics

- Attention mechanism

📘 Week 30: LLMs Introduction

- How GPT works

📘 Week 31: Using APIs

- OpenAI / other APIs

📘 Week 32: Build AI Apps

- Chatbots

📘 Week 33: Fine-Tuning Models

- Custom datasets

📘 Week 34: Vector Databases

- Embeddings

📓 Week 35: AI Deployment

- Flask / FastAPI

📓 Week 36: AI Project

👉 Build:

- AI-powered app (chatbot/tool)

⍰ PHASE 5: Tools + MLOps

📅 Weeks 37–44

📓 Week 37–38: Git & GitHub

- Version control

📓 Week 39–40: Docker

- Containerization

📓 Week 41–42: Cloud (AWS/GCP)

- Deploy models

📓 Week 43–44: MLOps Basics

- CI/CD for ML
- Model monitoring

🟢 PHASE 6: Projects (Portfolio)

📅 Weeks 45–50

📘 Week 45–46: Beginner Projects

- Spam classifier
- Simple prediction model

📘 Week 47–48: Intermediate Projects

- Image classifier
- NLP app

📘 Week 49–50: Advanced Project

👉 Build:

- AI SaaS tool
  - Chatbot
  - Recommendation system

🟦 PHASE 7: Job Preparation

📅 Weeks 51–52

📘 Week 51: Resume + Portfolio

- GitHub projects
- Kaggle profile

📓 Week 52: Apply + Interview Prep

- ML questions
- Case studies

🕒 Weekly Routine

- 5 days learning
- 2 days project/practice

Daily:

- 3–5 hours (AI needs more time)

🎯 Final Outcome (After 1 Year)

By Week 52:

✅ Strong Python + Math
✅ Machine Learning mastery
✅ Deep Learning basics
✅ NLP + AI apps
✅ 4–6 real projects
✅ Portfolio ready

👉 Job Roles:

- AI Engineer
- ML Engineer
- Data Scientist